Composition Cluster B1.03 + B1.04 + B1.05 — The Co-Required Entity Cluster Where Cells Provide Specific Operational Capabilities That Aspects Coordinate Into Purposeful Outcomes That Selves Integrate Into Coherent Intelligence, Each Entity Type Requiring the Other Two for Full Architectural Purpose

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Attribution

> This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, as a composition cluster, the co-required relationship among the three entity types — cell (B1.03), aspect (B1.04), and Self (B1.05) — established in the source paper's structural framework (§5.2), and to state the architectural consequences of that co-requirement across all three pairwise dependencies and the cluster as a whole.

Abstract

The source paper's structural framework (§5.2) introduces three entity types that populate the architecture's three levels: the cell as the atomic unit executing specific informational tasks, the aspect as the coordination arrangement of cells organized for a purpose, and the Self as the integrated whole holding multiple aspects as facets of one governed intelligence. This note formalizes the structural relationship among these three entity types as a co-required composition cluster. The co-requirement is precise: each entity type achieves its full architectural purpose only when the other two are present. Cells without aspects have no coordination governance layer to give their outputs direction; aspects without cells have nothing to coordinate; Selves without aspects have nothing to integrate at coordination scope; aspects without Selves have no integration intelligence architecture. The cell-Self dependency is mediated through aspects and the expression mechanism (B1.07) but is real: cells are the ultimate operational units whose aggregated execution constitutes the Self's intelligence, and Self governance reaches cells through that same mediation. The note states the three pairwise dependencies, the cluster-level co-presence requirement and what breaks when any entity type is absent, the complete architectural purpose statement that only the full cluster supports, the connection to recursive governance (B1.20), and the detection tests and failure modes for cluster incompleteness.

1. Why the three entity types require formalization as a composition cluster

The source paper introduces cell, aspect, and Self as three distinct levels in the architecture's structural framework (§5.2). The framework does not present them as independent additions to an otherwise complete architecture; it presents them as a structured composition in which the three levels mutually constitute one another. The aspect's definition presupposes cells as its content domain; the Self's definition presupposes aspects as its facets; cells' architectural purpose is realized through the coordination and integration layers the other two provide. The dependency runs in every direction.

Formalizing this as a composition cluster does work that the source paper's sequential introduction does not do alone. Sequential introduction names each entity type and its role; composition cluster formalization names the co-requirement — that each entity type requires the other two for full architectural purpose — and makes the consequences explicit. The consequences include three pairwise dependencies with precise directionality, a cluster-level co-presence requirement stronger than any individual pairwise dependency, and named failure modes for deployments that populate some levels but not all three. A practitioner who has read the source paper's structural framework knows what the three levels are; this note provides the framework for reasoning about what is missing when any one is absent.

The cluster also provides the entity-type population that the architecture's recursive governance commitment (B1.20) governs differentially across levels. Without all three entity types present, recursive governance has no three-level entity-type differentiation to bind its governance differentiation against. The cluster is therefore prerequisite to B1.20 operating as the source paper specifies.

2. Cluster identification: B1.03 + B1.04 + B1.05 as co-required entity types

Three entity types and their architectural roles, per §5.2 of the source paper:

B1.03 — Cell. The atomic unit from Paper 1 — a CKS artifact with substrates, orchestration rules, and the architectural commitments Paper 1 defends. All Paper 1 commitments hold at the cell level without modification. The cell handles a specific informational task; its scope is bounded; its substrates carry the DNA and action layers that define its potential and record its experience. The cell is the architecture's *operational execution unit*: the level at which work is done, outputs are produced, and the intelligence that higher levels coordinate and integrate originates.

B1.04 — Aspect. A coordination arrangement of cells serving a particular purpose. The aspect operates over its constituent cells as its content domain, asking pattern questions across them in service of its purpose. Multiple aspects can coexist within one Self, with cells participating in multiple aspects through relational role membership. The aspect is the architecture's *coordination governance unit*: the level at which cells' individual execution outputs are organized into purposeful coordination outcomes rather than remaining as isolated task results.

B1.05 — Self. The integrated whole that holds multiple aspects as facets of one CKS-governed intelligence. The Self is not a bigger aspect, not a rigid top-of-hierarchy, and not one end of a strict access hierarchy. The Self contains the multiple coexisting structural perspectives and their collective intelligence as one unified whole. The Self is the architecture's *integration governance unit*: the level at which coordination-governed outputs from multiple aspects are held under unified integration intelligence, producing coherent AI governed at the scope of the whole.

The relationship structure is directional in both the downward and upward sense. Each level's definition names the level below as the substance it operates over: the aspect's definition names cells as its content domain; the Self's definition names aspects as its facets. And each level's definition implies the level above as providing the governance architecture that gives its outputs purpose beyond the level itself. The three are co-required.

3. Pairwise dependency: B1.03 + B1.04 (execution-coordination)

The first pairwise dependency holds between the cell level and the aspect level.

Cells provide execution substance that aspects require. An aspect is defined as a coordination arrangement of cells. Without cells providing operational execution capability, an aspect has coordination architecture with nothing to coordinate. The aspect can specify its purpose, its coordination patterns, its relational role assignments — but all of those specifications are vacuous absent cells to which they apply. An aspect without cells is a coordination governance structure that governs nothing; its role as coordination unit is uninstantiated.

Aspects provide coordination governance that cells require. A cell executing without aspect-level coordination governance operates in informational isolation. Its outputs accumulate as task executions, but no coordination governance layer determines how those outputs combine, what patterns they contribute to, or what purpose organizes their relationship to one another. Cells without aspects produce isolated informational task execution — the work is performed but in coordination vacuum. The architectural function of the aspect — organizing execution into purposeful coordination — is absent, and no amount of cell-level sophistication recovers it, because coordination governance is not a property the cell can provide for itself.

The pairwise dependency: cells provide the execution substance that makes aspects' coordination governance operative; aspects provide the coordination governance purpose that makes cells' execution architecturally meaningful beyond the isolated task. Each requires the other for full architectural purpose.

4. Pairwise dependency: B1.04 + B1.05 (coordination-integration)

The second pairwise dependency holds between the aspect level and the Self level.

Aspects provide coordination-governed substance that Selves require. A Self is defined as the integrated whole holding multiple aspects as facets of one intelligence. Without aspects providing coordination-organized outputs, the Self has integration architecture with nothing to integrate at the coordination-organized level. The Self can access cells directly when purpose requires (§5.2), but the Self's integration of aspects is what makes multiple coordination arrangements cohere as one intelligence rather than as parallel operational activity. Selves without aspects lack the coordination-level substance that the Self's integration architecture is defined as integrating.

Selves provide integration intelligence that aspects require. Aspects operating without Self-level integration governance have no architecture determining how multiple aspects' coordination outcomes relate to one another and cohere as unified intelligence. Multiple aspects producing coordination-governed outputs in parallel do not automatically produce integrated intelligence; they produce parallel coordination. The Unintegrated Self anti-pattern (B3.06) names this condition: aspects exist but the Self-level integration function that makes their coexistence coherent governance is absent. Each aspect operates without the integration architecture that gives its coordination purpose meaning within the whole.

The pairwise dependency: aspects provide the coordination-governed substance that Self-level integration requires for coherent synthesis; Selves provide the integration intelligence that makes aspects' coordination outcomes cohere as unified governed intelligence rather than as independent parallel outputs. Each requires the other for full architectural purpose.

5. Pairwise dependency: B1.03 + B1.05 (execution-integration)

The third pairwise dependency holds between the cell level and the Self level. This dependency is mediated — aspects and the expression mechanism (B1.07) are the structural path through which it operates — but it is architecturally real, not merely a corollary of the first two pairwise dependencies.

Cells are the ultimate operational units whose aggregated execution constitutes the Self's intelligence. The Self's coherent intelligence is not generated independently of its cells; it is what the combined capabilities of the cells, coordinated through aspects and integrated through the Self's governance, produce. Remove the cells and the Self's integration architecture has no operational execution providing the raw intelligence it integrates. The chain runs through aspects, but the terminus on the operational side is cell execution. What the Self ultimately integrates is what cells do.

Self governance reaches cells through expression and coordination. The expression mechanism (B1.07) selects which DNA-layer substrates activate for a given cell goal, and that expression is itself human-governed and shaped by Self-level governance context. Aspect coordination rules govern how cells participate in aspect-level activities. Both mechanisms carry Self-level governance influence to cell-level operation. Cells without Self have no integration context for their outputs; Self without cells has no execution providing the operational substance that integration governs.

The pairwise dependency is mediated but real: cells are the ultimate operational units of the Self's intelligence; Self governance ultimately shapes cells through the expression mechanism and through aspect coordination; neither cell scope nor Self scope achieves its architectural purpose without the other.

6. The architectural purpose statement and cluster co-presence requirement

The complete cluster B1.03 + B1.04 + B1.05 produces an architecture with three mutually required dimensions:

Specific capabilities at the cell level: atomic operational execution of specific informational tasks, each under Paper 1's commitments, each carrying DNA and action layers, each subject to governed lifecycle.

Purposeful organization at the aspect level: coordination governance that organizes cells' execution capabilities into purposeful coordination outcomes, giving execution outputs direction and relational meaning beyond the isolated task.

Coherent integration at the Self level: integration intelligence that holds multiple aspects' coordination outcomes as facets of one governed intelligence, providing the unified whole at which the architecture's instinct/reasoning separation operates at full scope.

Each dimension is irreplaceable. Specific capabilities without purposeful organization and coherent integration produce isolated execution. Purposeful organization without specific capabilities and coherent integration produces coordination architecture without operational substance and without integration. Coherent integration without specific capabilities and purposeful organization produces integration architecture without the substance it is defined to integrate.

The cluster co-presence requirement follows: the architecture is complete only when all three entity types are present and operative.

B1.03 + B1.04 without B1.05 (cells and aspects, no Self): purposefully coordinated capabilities exist but are not integrated into coherent intelligence. Multiple aspects operate in parallel without integration governance determining how their coordination outcomes cohere. The system produces coordination-organized outputs per aspect but no integration architecture makes those outputs cohere as one governed intelligence. The Unintegrated Self anti-pattern (B3.06) names this condition when it arises from incomplete cluster instantiation.

B1.03 + B1.05 without B1.04 (cells and Self, no aspects): cells execute and the Self holds architecture for integration, but the aspect coordination layer through which aspect-level coordination governance operates is absent. The three-level structure §5.2 establishes is populated at two levels; the aspect layer that the Self's integration is defined as integrating does not exist. The Flat Architecture anti-pattern (B3.03) names the related structural failure when the aspect layer is missing, collapsing the three-level architecture to a functionally two-level structure.

B1.04 + B1.05 without B1.03 (aspects and Self, no cells): coordination and integration governance architecture exists but is operationally empty. The aspect cannot coordinate cells that do not exist; the Self cannot integrate aspect-level outputs that aspects cannot produce without cells. Coordination and integration structure is instantiated without execution.

7. Recursive governance connection

The source paper's recursive governance commitment (B1.20) applies at each of the three entity-type levels: governance-boundary inheritance means that Paper 1's architectural commitments hold at the cell level without modification, and that governance at the aspect and Self levels applies the same commitments at the scope appropriate to each level. Recursive governance differentiates its governance scope according to which entity type it governs: cell-level governance applies to atomic task execution; aspect-level governance applies to coordination arrangements; Self-level governance applies to integration of the whole.

The B1.03 + B1.04 + B1.05 cluster provides the entity-type population that B1.20 governs differentially. Without the cluster, recursive governance has no three-level entity-type differentiation against which to scope its differentiated governance. A deployment with only cells has governance at cell scope only — the recursive extension to aspect scope and Self scope has no entity types to bind to. A deployment with cells and aspects but no Self has governance differentiation at two levels but not the integration scope at which the Self's unified governance applies. The complete cluster is prerequisite to B1.20 operating as the source paper specifies.

The recursive connection runs in both directions: the cluster provides the entity-type structure that B1.20 requires; B1.20's governance differentiation is what makes the cluster's three levels architecturally coherent under unified governance rather than three independently operating tiers with accidental level structure.

8. Operational integration, detection, and failure modes

Operational integration. In a complete cluster deployment, the three entity types are operationally interdependent across the lifecycle. Cells are originated with specific informational task scope under the birth mechanism; aspects group cells under coordination purpose from origination; Selves hold aspects as integration facets from Self origination. During operation: cells execute their informational tasks, producing outputs that their aspects draw on in asking coordination-pattern questions; aspects coordinate, producing coordination-organized information that informs Self-level integration; Selves govern integration, shaping which aspects are active and how they relate, shaping aspects' coordination patterns, which shapes cells' expression activation through the expression mechanism (B1.07). The governance flow — Self shaping aspects, aspects shaping cells, cells informing aspects, aspects informing Self — is the operational architecture when the complete cluster is present and functioning.

Detection. A deployment instantiates the complete cluster if and only if: (1) *Entity-level distinguishability* (B2.09) — entities at all three levels exist and are distinguishable as cells (atomic task execution scope), aspects (coordination-purpose scope), and Self (integration scope); a deployment in which all entities operate at effectively the same scope fails this test. (2) *Level-specific inheritance verification* (per B2.14, B2.19, B2.24) — entities at each level satisfy the level-appropriate commitments; cell-level entities satisfy Paper 1's commitments without modification; aspect-level entities satisfy the coordination-arrangement commitments of B1.04; Self-level entities satisfy the integration-whole commitments of B1.05. (3) *Cluster completeness* — the deployment has entities actually operative at all three levels, not merely claimed levels without populated entity types.

Failure modes. Four named failure conditions apply to cluster-incomplete deployments: the *Flat Architecture* anti-pattern (B3.03) for structural cluster failure in which the aspect layer is missing and the three-level architecture collapses; the *Undifferentiated Cell* anti-pattern (B3.04) for cell-level entity failure in which a cell's scope is not bounded to a specific informational task and takes on aspect-level or Self-level work; the *Purposeless Aspect* anti-pattern (B3.05) for aspect-level entity failure in which an aspect exists without a defined coordination purpose and does not function as coordination governance; and the *Unintegrated Self* anti-pattern (B3.06) for Self-level entity failure in which aspects exist but are not held under Self-level integration governance.

9. Conclusion

The cell, aspect, and Self are not three independently useful entity types that happen to appear in the same architecture. They are co-required: each entity type achieves its full architectural purpose only when the other two are present. Cells provide specific execution capabilities that aspects coordinate into purposeful outcomes that Selves integrate into coherent governed intelligence. The dependency runs through all three pairwise relationships — execution-coordination (B1.03 + B1.04), coordination-integration (B1.04 + B1.05), and the mediated but real execution-integration (B1.03 + B1.05) — and the cluster is architecturally complete only when all three entity types are populated and operative.

The cluster's architectural purpose — specific capabilities, purposefully organized, coherently integrated — is the entity-type instantiation of the three-level structure §5.2 of the source paper establishes. Formalizing it as a composition cluster makes the co-requirement explicit as a constraint on deployment: a system that populates two of the three entity types but not the third is architecturally incomplete in a specific and named way. The failure modes (B3.03 through B3.06) name the incomplete configurations; this note names the complete one and the structural requirement that completeness places on any CKS deployment operating at multi-level scope.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

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